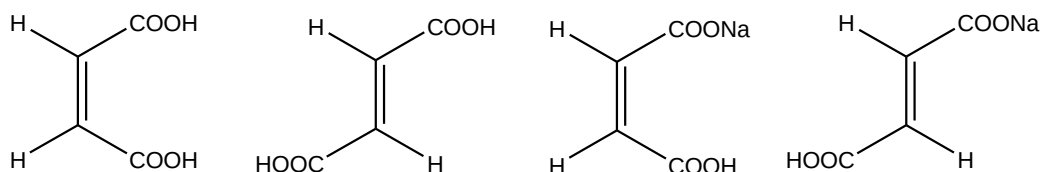


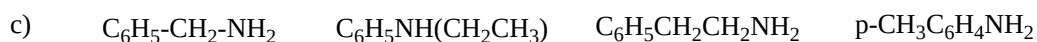
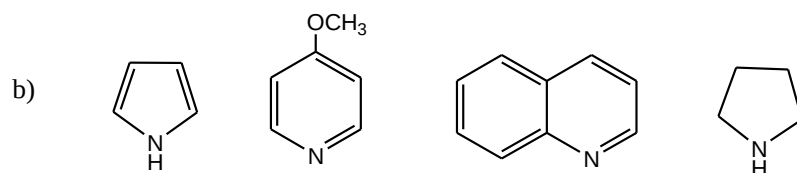
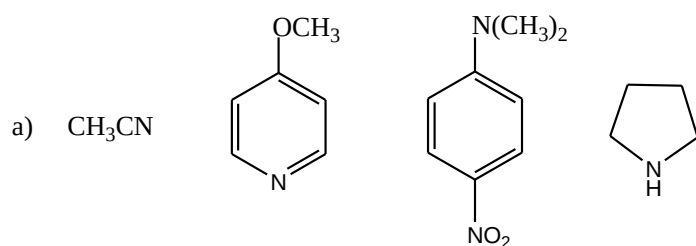
## ACML100 Tutorial 3

### Determination of reaction mechanism and Organic acids and bases

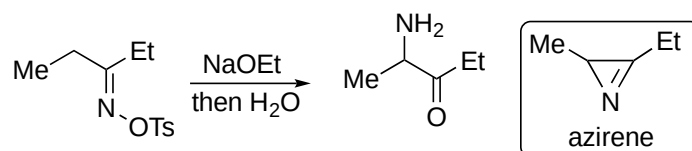
1. (a) Explain the trend observed in the pK<sub>a</sub> of methanoic acid (pK<sub>a</sub> 3.75), phenyl ethanoic acid (pK<sub>a</sub> 4.31), and ethanoic acid (pK<sub>a</sub> 4.76).  
  
(b) Explain the observed pK<sub>a</sub> of various *p*-substituted benzoic acids. X=CN(3.54), OH(4.58), OCH<sub>3</sub>(4.47), CH<sub>3</sub>(4.31), NO<sub>2</sub>(3.43), H(4.17).  
  
(c) Arrange the following acids in expected order of increasing pK<sub>a</sub> values:  
(i) CH<sub>3</sub>COOH, (CH<sub>3</sub>)<sub>3</sub>N<sup>+</sup>CH<sub>2</sub>COOH, HOOCCH<sub>2</sub>COONa, O<sub>2</sub>NCH<sub>2</sub>COOH.  
(ii) acetic acid, cyanoacetic acid, hydroxyl acetic acid, nitroacetic acid.  
  
(d) Why do the dissociation constants of substituted benzoic acids vary as given below?  
*p*-methoxybenzoic acid ( $3.38 \times 10^{-5}$ ), *m*-methoxybenzoic acid ( $8.17 \times 10^{-5}$ ), benzoic acid ( $6.27 \times 10^{-5}$ ).  
  
(e) Why is 2,4,6-trinitro-N,N,-dimethylaniline about 40,000 times stronger base than 2,4,6-trinitroaniline.  
  
(f) For the following two pairs of acids, indicate which is the stronger acid in each pair and why?



2. Arrange the following in the increasing order of their relative basicity.



3. For the following transformation, intermediate azirene was isolated. With the help of intermediate structure, propose a suitable mechanism for this reaction.



4. Explain the mechanism of the following transformation and identify the rate determining step.

